

Clinical factors predictive of tumour cytogenetics and metastasis in 86 patients with choroidal nevus growth into melanoma (DIP and DOT study)

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ABSTRACT

Background Choroidal melanoma can arise from malignant transformation of choroidal nevus. The Cancer Genome Atlas, a classification system based on the genetic status of chromosomes 3 and 8, can be used to prognosticate metastasis and death in choroidal melanoma. This study explores the impact that growth rates and clinical/imaging (TFSOM-DIM (To Find Small Ocular Melanoma, Doing IMaging)) risk factors have on cytogenetics, metastasis and death in choroidal nevus which transforms into melanoma.

Methods A retrospective study was performed on 86 consecutive patients diagnosed with choroidal nevus with transformation into melanoma. Tumour cytogenetic results and TFSOM-DIM risk factors for transformation at the date of initial presentation (DIP) and date of transformation diagnosis (DOT) were recorded.

Cytogenetic testing of tumours was offered to patients at DOT and was performed on tumour samples obtained via fine-needle aspiration biopsy prior to plaque radiotherapy initiation.

Results Of 86 patients, 66% were cytogenetically low-risk and 34% were cytogenetically high-risk. At DOT, high-risk tumours possessed more TFSOM-DIM risk factors than low-risk tumours (4.5 vs 3.8, $p=0.003$). Choroidal nevus with growth in thickness >0.5 mm/year or $>20\%$ per year had increased risk for high-risk cytogenetics (relative risk (RR)=1.93, 95% CI 1.09 to 3.43, $p=0.027$; RR=2.18, 95% CI 1.22 to 3.92, $p=0.008$, respectively). Choroidal nevus with growth in basal diameter >0.7 mm/year or $>10\%$ per year had increased risk for high-risk cytogenetics (RR=2.22, 95% CI 1.25 to 3.93, $p=0.007$; RR=1.83, 95% CI 1.03 to 3.26, $p=0.042$, respectively).

Conclusions When monitoring patients with choroidal nevus, clinical risk factors are important in estimating risk for transformation into melanoma. We found that thickness growth >0.5 mm/year or $>20\%$ per year, as well as basal diameter growth >0.7 mm/year or $>10\%$ per year, were all thresholds that demonstrated increased risk for high-risk cytogenetics. Additionally, tumours with a greater number of TFSOM-DIM risk factors by DOT had significantly increased likelihood of high-risk cytogenetics.

INTRODUCTION

Choroidal melanoma can arise from malignant transformation of benign choroidal nevus.¹ This

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Benign choroidal nevus carries the risk for transformation into malignant choroidal melanoma, which can be estimated by the To Find Small Ocular Melanoma, Doing IMaging (TFSOM-DIM) clinical and imaging risk criteria. Choroidal melanoma carries the potential for metastasis and death, which is routinely prognosticated via cytogenetic testing.

WHAT THIS STUDY ADDS

⇒ This study shows that clinical and imaging risk factors as well as rates of growth for a choroidal nevus can inform on the tumour's cytogenetic properties on transformation into malignant melanoma.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Cytogenetic testing is the gold standard for prognosticating choroidal nevus transformation to melanoma, but in cases where such testing is not performed, TFSOM-DIM clinical and imaging criteria and choroidal nevus growth rates may offer crucial information into the likelihood of favourable or unfavourable prognostic cytogenetic characteristics and metastatic outcomes.

malignancy poses a risk for metastasis, with the most common sites of metastasis being liver, lung, bone and skin, respectively.² Genetic testing of choroidal melanoma is routinely performed for DNA or RNA prognostication of risk for metastatic disease. The Cancer Genome Atlas (TCGA) is a genetic classification system for DNA or RNA data and stratifies tumours based on increasing risk for metastasis.^{3–6} Robertson *et al* performed a multiplatform genetic analysis on 80 eyes with uveal melanoma as part of the TCGA project supported by the National Cancer Institute Center for Cancer Genomics and the National Human Genome Research Institute and extrapolated numerous molecular pathways involved in the pathogenesis of choroidal melanoma.³ This analysis was later synthesised by Jager *et al* into a 4-class prognostication system based on chromosomal aberrations, consisting of TCGA Group A (disomy 3, no 8q gain), TCGA Group B

(disomy 3, partial 8q gain), TCGA Group C (monosomy 3, 8q gain) and TCGA Group D (monosomy 3, multiple 8q gains).⁴ The TCGA classification system has since been applied to a cohort of 658 uveal melanoma patients and later 1001 uveal melanoma patients, by Shields *et al*, demonstrating its accuracy and reliability as a predictive tool for risk for melanoma-related metastasis and death.^{5,6}

In an analysis of 3806 patients with choroidal nevus, six risk factors on multimodal imaging were associated with the potential for growth of nevus into melanoma.⁷ These factors are remembered by the mnemonic TFSOM-DIM (To Find Small Ocular Melanoma, Doing Imaging) representing thickness >2 mm on ultrasonography, presence of subretinal fluid (SRF) on optical coherence tomography (OCT), visual symptoms defined as visual acuity of 20/50 or worse by Snellen acuity, the presence of orange pigment on fundus autofluorescence, melanoma hollow-ness on ultrasonography and basal diameter >5 mm by fundus photography.⁷ Risk for the growth of a nevus into melanoma was found to be 11% at 5 years with one risk factor and increased to greater than 50% with four or more risk factors.⁷ While these clinical factors portend a risk for transformation, their individual and collective correlation with tumours within each TCGA group, and therefore risk of metastasis and death from choroidal melanoma, has not been established.

Prior to the introduction of the TCGA prognostication system and the updated TFSOM-DIM clinical/imaging criteria, a study of 55 patients with choroidal nevus transformation into melanoma suggested a correlation between genetic aberrations and rate of tumour growth.³⁻⁸ Slow growth (nevus transformation into melanoma in >1 year) represented low-risk tumours with fewer mutations on chromosome 3 or 8, whereas fast growth (nevus transformation into melanoma in ≤ 1 year) represented higher-risk tumours with increased aberrations on chromosomes 3, 8 or both.⁸

While choroidal nevus carries a low risk of transformation into melanoma, the sight-threatening and life-threatening nature of choroidal melanoma makes persistent surveillance paramount.⁹ The utilisation of these clinical risk factors for predicting the likelihood of transformation may inform the clinician when to adjust the follow-up interval or initiate treatment in cases where suspicion of transformation is high.^{10,11} However, the utility of these specific clinical features for estimating outcomes of TCGA cytogenetic analysis, the primary tool for prognosticating metastasis and death, has not been explored. Our group previously evaluated the rate of transformation of choroidal nevus into melanoma as a predictor of cytogenetic profile, with tumours transforming into melanoma in ≤ 1 year from DIP having a higher risk of genetic aberrations on chromosomes 3, 8 or both.⁷ Relying on such time intervals may be difficult in cases where the patient was not followed up regularly and the date of initial onset of choroidal nevus is not possible to determine. Therefore, we sought to explore the predictive value of objective measures of choroidal nevus growth, namely rate of increase in thickness and rate of increase in basal diameter.

Here, we specifically assess patients with previously diagnosed choroidal nevus that demonstrated evidence of growth into choroidal melanoma and study the DNA mutation(s) within the melanoma. With the advent of the refined TCGA tumour cytogenetic classification system, this study of a large pool of patients with choroidal nevus growth into melanoma, comparing clinical risk factors, TCGA cytogenetics and patient outcomes, provides novel insight into using clinical factors observed prior to transformation to predict tumour cytogenetics which informs on prognostic outcomes. We hypothesise that choroidal nevus with

a greater rate of growth in base and thickness potentially demonstrates higher-risk melanoma on TCGA DNA testing (TCGA Groups C or D), compared with choroidal nevus with a slower growth rate in those with lower-risk melanoma (TCGA Groups A or B).

METHODS

A retrospective review was performed on patients diagnosed with choroidal nevus with transformation into melanoma, confirmed on the ocular oncology service at Wills Eye Hospital, Philadelphia, Pennsylvania, USA, between 18 March 1988 and 13 December 2022. All patients were evaluated by the senior author (CLS) using a multimodal approach consisting of clinical and photographic evaluation methods, including slit-lamp biomicroscopy, indirect ophthalmoscopy, fundus photography, OCT and B-scan ultrasonography. The same clinical and photographic evaluation methods were performed at each follow-up visit, and the diagnosis of nevus transformation to melanoma was made if there were significant differences in any individual or combination of risk factors that were suspected to be caused by malignant tumour activity. Patients were included if there was photographic documentation of growth into melanoma and they agreed to tumour cytogenetic testing.

The diagnosis of melanoma and subsequent cytogenetic testing occurred between 13 December 2005 and 16 August 2022. The patients were assessed for demographic features including age, sex, race, affected eye and follow-up interval. Clinical features were collected from the date of initial presentation (DIP), when the patient's nevus was first identified at the ocular oncology centre at Wills Eye Hospital and date of transformation diagnosis (DOT), when the patient was diagnosed with melanoma. These clinical features included tumour location, basal diameter in mm assessed by fundus photography, thickness (mm) measured by ultrasonography, presence of any clinically evident SRF assessed by OCT, presence of visual symptoms defined as visual acuity of 20/50 or worse by Snellen testing, presence of any hyperautofluorescence (lipofuscin) on fundus autofluorescence suggesting orange pigment and tumour acoustic features on ultrasonography (acoustic hollowness evident on A-scan and B-scan). For patient visits prior to 2002, when our team acquired access to OCT technology, SRF on fundus photography was used as a surrogate (10 DIPs, zero DOTs). For patient visits prior to 2007 when our team acquired access to autofluorescence technology, orange pigment on fundus photography was used as a surrogate (23 DIPs, 6 DOTs).

The rate of growth of each nevus to melanoma (time elapsed between DIP and DOT as well as average rate of base size and thickness increase) and the TCGA genetic outcome were recorded for each case. Data on outcomes of melanoma-related metastasis and death were collected and verified from patient medical records from medical oncologists, family correspondences and death certificates, when available. Tumours in TCGA groups A or B were designated as low-risk and tumours in TCGA groups C or D were designated as high-risk.

Fine-needle aspiration was performed in the operating room immediately prior to plaque radiotherapy application. Samples were collected either through a transscleral approach through the pars plana or tumour base using a 27-gauge needle attached to a 10-inch tube and a 10 cc syringe. Cellular samples were stored in Hank's solution (Gibco, Life Technologies, Grand Island, New York, USA) at 4°C in preparation for cytogenetic analysis with a DNA Micro Kit (QIAGEN, Valencia, California, USA).

Statistical analysis

Data were collected using Microsoft Excel (V.16.66.1, Redmond, Washington, USA). A paired t-test was used to compare paired continuous variables from different time points and χ^2 test was used for categorical variables. Univariate and multivariate analyses were performed when indicated using R Statistical Software V.4.0.3 (Foundation for Statistical Computing, Vienna, Austria) and RStudio software V.2022.02.3.492 (RStudio Team, Boston, Massachusetts, USA). Variables found to be significant at the level of p value <0.10 were included in the multivariate regression model. Kaplan-Meier analyses for outcomes of metastasis and death from melanoma were also performed using R Statistical Software V.4.0.3. Statistical significance was defined as p value <0.05.

RESULTS

There were 86 patients with choroidal nevus that demonstrated transformation into melanoma and who had cytogenetic testing over an overall diagnosis and observation period of 34 years, and a transformation period of 16 years. There were 80 (93%) patients who demonstrated nevus growth in thickness measured by B-scan ultrasonography and 71 (83%) patients who demonstrated growth in basal dimension measured by colour fundus photography. Notably, 67 (78%) patients demonstrated growth in both thickness and basal dimension. All patients were stratified by cytogenetic outcome, with 57 (66%) patients having cytogenetic results of TCGA A or B, and 29 (33%) patients having cytogenetic results of TCGA C or D. Among the low-risk cytogenetics group, the mean absolute growth in tumour thickness was 0.5 mm/year (median 0.2 mm/year, range 0–4.8 mm/year), and the mean percentage growth in tumour thickness was 22% per year (median 12% per year, range 0–178% per year). Among the same low-risk cytogenetics group, the mean absolute growth in tumour basal diameter was 0.7 mm/year (median 0.3 mm/year, range 0–12.0 mm/year), and the mean percentage growth in tumour basal diameter was 13% per year (median 5% per year, range 0–200% per year). Among the high-risk cytogenetics group, the mean absolute growth in tumour thickness was 1.1 mm/year (median 0.5 mm/year, range 0–11.6 mm/year), and the mean percentage growth in tumour thickness was 44% per year (median 23% per year, range 0–446% per year). Among the same high-risk cytogenetics group, the mean absolute growth in tumour basal dimension was 2.3 mm/year (median 0.7 mm/year, range 0–32.0 mm/year), and the mean percentage growth in tumour basal diameter was 33% per year (median 9% per year, range 0–400% per year).

The mean rates of absolute tumour growth in thickness (increase of 0.5 mm/year) and in base size (increase of 0.7 mm/year) in the low-risk cytogenetics group were used as reference points for comparison and acted as the limits for defining slow rate of growth versus fast rate of growth in their respective dimensions. The rate of absolute growth in tumour thickness increase was ≤ 0.5 mm/year in 58 (67%) patients and >0.5 mm/year in 28 (33%) patients. The rate of absolute growth in tumour basal diameter was ≤ 0.7 mm/year in 58 (67%) patients and >0.7 mm/year in 28 (33%) patients. Similarly, the mean rates of percent growth in the low-risk cytogenetics group rounded to the nearest tens place (percent thickness increase of 20% per year and percent base increase of 10% per year) were used as reference points for comparison and acted as the limits for defining slow rate of growth versus fast rate of growth. The rate of tumour thickness percent increase was $\leq 20\%$ per year in 55 (64%) patients and $>20\%$ per year in 31 (36%) patients.

Table 1 Clinical factors predictive of tumour cytogenetics in 86 patients with choroidal nevus growth into melanoma: patient demographics

Demographic features n=86(n (%))	
Age at presentation (years)	
Mean (median, range)	58 (59, 10–83)
Sex	
Male	36 (41.9)
Female	50 (58.1)
Race	
Caucasian	85 (98.8)
African American	0 (0)
Hispanic	0 (0)
Asian	1 (1.2)
Other	0 (0)
Unknown	0 (0)
Affected eye	
Right	40 (46.5)
Left	46 (53.5)
Follow-up intervals	
Months from DIP to DOT mean (median, range)	57.7 (44, 2–313)
Months from DIP to date last seen mean (median, range)	120.2 (107.5, 3–335)

DIP, date of initial presentation; DOT, date of transformation.

The rate of tumour basal diameter percent increase was $\leq 10\%$ per year in 57 (66%) patients and $>10\%$ per year in 29 (34%) patients.

Patient demographical data are shown in table 1 with anatomical features of choroidal tumors shown in online supplemental table 1. The mean age of the patients was 58 years old and there was a predominance of White patients, with more females than males. Mean interval of follow-up of choroidal nevi at risk (time from DIP to DOT) was 57.7 months and mean total follow-up interval was 120.2 months. The tumor anterior margin was most commonly between the macula (anatomic macula defined as the area confined by the superior and inferior arcades, optic nerve and 2 disc diameters temporal to central fovea) and equator (n=46, 53.5%). The posterior tumor margin was most commonly in the macula (n=44, 51.2%).

Clinical features of choroidal nevus are shown in table 2. At DIP, mean tumour thickness was 2.2 mm and mean tumour basal diameter was 7.4 mm. At DOT, mean tumour thickness was 3.5 mm (59.1% increase, $p<0.0001$) and mean tumour basal diameter was 9.8 mm (32.4% increase, $p<0.00001$). TFSOM-DIM risk factors for transformation into melanoma reveal differences at DIP and DOT in all six criteria. The mean number of TFSOM-DIM risk factors present at DIP was 2.1 and increased by DOT to 4.1 ($p<0.0001$).

A comparison of clinical characteristics and TFSOM-DIM criteria at DIP and DOT in tumours with low-risk (vs high-risk) cytogenetics at DOT are shown in table 3. The clinical features at DIP that are predictive of high-risk cytogenetics at DOT included presence of subretinal fluid on OCT (relative risk (RR)=2.62, 95% CI 1.25 to 5.49, $p=0.015$). The feature predictive of high-risk cytogenetics at DOT was also the presence of subretinal fluid on OCT (RR=1.36, 95% CI 1.10 to 1.82, $p=0.038$). Tumours with high-risk cytogenetics demonstrated no difference in number of TFSOM-DIM risk factors at DIP, but by DOT demonstrated greater number of TFSOM-DIM risk criteria (4.5 vs 3.8, $p=0.003$).

Table 2 Clinical factors predictive of tumour cytogenetics in 86 patients with choroidal nevus growth into melanoma: clinical features and risk factors (TFSOM-DIM criteria*) of all tumours at date of initial presentation (DIP) and date of transformation (DOT) into melanoma

Clinical features of the tumour n=86(n (%))				
Tumour feature	Choroidal nevus at date of initial presentation (DIP)	Choroidal melanoma at date of transformation (DOT)	Percent change (%)	P value
Thickness (mm) mean (median, range)	2.2 (2.1, 0.5–4.3)	3.5 (3.4, 1.3–8.7)	59.1	p<0.00001
Diameter (mm) mean (median, range)	7.4 (7.0, 1.5–18.0)	9.8 (9.0, 5.0–19.0)	32.4	p<0.00001
Risk factors related to transformation into melanoma (TFSOM-DIM criteria*)				
All patients n=86(n (%))				
Thickness>2 mm by ultrasonography	46 (53.5)	77 (89.5)	36.0	p<0.00001
Presence of subretinal fluid on optical coherence tomography	21 (24.4)	61 (70.9)	46.5	p<0.00001
Symptoms of vision ≤20/50 by Snellen visual acuity testing	4 (4.7)	20 (23.3)	18.6	p<0.001
Presence of orange pigment by fundus autofluorescence	9 (10.5)	40 (46.5)	36.0	p<0.00001
Tumour hollowness by ultrasonography	41 (47.7)	69 (80.2)	32.5	p<0.00001
Diameter >5 mm on fundus photography	62 (72.1)	83 (96.5)	24.4	p=0.00001
Total TFSOM-DIM score mean (median, range)	2.1 (2.0, 0.0–6.0)	4.1 (4.0, 1.0–6.0)	91.3	p<0.00001
TFSOM-DIM Criteria, remembered by mnemonic 'To Find Small Ocular Melanoma, Doing IMaging': T: thickness >2 mm on ultrasonography; F: presence of subretinal fluid on spectral domain optical coherence tomography; S: visual symptoms of visual acuity of 20/50 or worse by Snellen testing; O: presence of orange pigment on fundus autofluorescence; M: Melanoma hollowness on ultrasonography; DIM: basal diameter greater than 5 mm by fundus photography.				
*TFSOM-DIM criteria adapted from: Shields CL, Dalvin LA, Ancona-Lezama D, <i>et al.</i> Choroidal nevus imaging features in 3,806 cases and risk factors for transformation into melanoma in 2,355 cases: The 2020 Taylor R. Smith and Victor T. Curtin Lecture. <i>Retina</i> 2019;39:1840–1851.				

A comparison of cytogenetic outcomes based on rates of tumour thickness increase and tumour basal diameter increase is shown in [table 4](#). Tumours with greater rates of absolute growth in thickness and basal dimension and those with greater rates of percent growth in thickness and basal dimension showed greater likelihood for high-risk cytogenetics. High-risk cytogenetics were found with absolute thickness increase >0.5 mm/year (26% vs 50%, RR=1.93, 95% CI 1.09 to 3.43, p=0.027) and with absolute basal diameter increase >0.7 mm/year (24% vs 54%, RR=2.22, 95% CI 1.25 to 3.93, p=0.007). Similarly, high-risk cytogenetics were found with percent thickness increase >20% per year (24% vs 52%, RR=2.18, 95% CI 1.22 to 3.92, p=0.008) and with percent basal diameter increase >10% per year (26% vs 48%, RR=1.83, 95% CI 1.03 to 3.26, p=0.042).

Kaplan-Meier analyses of outcomes of time to transformation diagnosis from nevus to melanoma, melanoma-related metastasis and melanoma-related death are shown in online supplemental table 2 and figures 1–4. There was no statistically significant difference in event-free survival of transformation from nevus to melanoma between those with low-risk cytogenetics and those with high-risk cytogenetics (online supplemental figure 1). Tumours possessing more TFSOM-DIM risk factors at DOT (1–3 vs 4 vs 5 vs 6) were significantly more likely to have melanoma-related metastasis by 3 years from DOT onward (0% vs 12% vs 8% vs 40%, p=0.04); however, post-hoc pairwise comparisons showed no significant difference between any individual groups. There was no statistically significant difference in melanoma-related death from DOT onward with different numbers of TFSOM-DIM criteria at DOT (online supplemental figure 2). There was no difference in melanoma-related metastasis or death from DOT onward between patients with thickness growth ≤ 0.5 mm/year vs >0.5 mm/year (online supplemental figure 3A,B), patients with thickness growth ≤ 20% per year vs >20% per year (online supplemental figure 4A,B), patients with basal dimension growth ≤ 0.7 mm/year vs >0.7 mm/year (online supplemental figure 3C,D) or patients with basal dimension growth ≤ 10% per year vs >10% per year (online supplemental figure 4C,D).

DISCUSSION

Here, we have shown that tumours with rates of absolute or percent increase in diameter or thickness above certain thresholds have increased risk for having high-risk cytogenetic profile. Specifically, an absolute increase in thickness >0.5 mm/year demonstrated a significantly higher rate of high-risk cytogenetics compared with tumours with an absolute increase in thickness ≤ 0.5 mm/year (RR=1.93, 95% CI 1.09 to 3.43, p=0.027, [table 3](#)). Regarding the rate of percent increase in thickness, tumours that grew in thickness >20% per year also demonstrated a significantly higher rate of high-risk cytogenetics compared with tumours that grew in thickness ≤ 20% per year (RR=2.18, 95% CI 1.22 to 3.92, p=0.008, [table 4](#)). Furthermore, tumours with absolute increase in basal diameter >0.7 mm/year also demonstrated a significantly higher rate of high-risk cytogenetics compared with tumours with absolute increase in basal diameter ≤ 0.7 mm/year (RR=2.22, 95% CI 1.25 to 3.93, p=0.007, [table 3](#)). Finally, tumours with percent increase in basal diameter >10% per year also demonstrated a significantly higher rate of high-risk cytogenetics compared with tumours with percent increase in basal diameter ≤ 10% per year (RR=1.83, 95% CI 1.03 to 3.26, p=0.042, [table 4](#)).

The only TFSOM-DIM risk factor at both DIP and DOT that was predictive of high-risk cytogenetics at DOT was the presence of subretinal fluid on OCT (RR=2.62, 95% CI 1.25 to 5.49, p=0.015 at DIP; RR=1.36, 95% CI 1.10 to 1.82, p=0.038 at DOT; [table 3](#)). While this is similar to previously reported findings, the pathogenesis of this significance remains unclear.^{12 13} One study speculated that the presence of subretinal fluid may be due to extravasation caused by choriocapillaris compression as the nevus grows, making it a result of the growth itself rather than an independent risk factor.^{14 15} Choroidal nevi that have transformed into melanoma with high-risk cytogenetics (TCGA C and D) possess significantly more TFSOM-DIM risk factors at DOT than tumours with low-risk cytogenetics (4.5 vs 3.8, p=0.003). This finding is further strengthened as patients with tumours that have more TFSOM-DIM criteria by DOT are significantly more likely to have been diagnosed with metastatic disease by 3 years after DOT (p=0.04, online supplemental

Table 3 Clinical factors predictive of tumour cytogenetics in 86 patients with choroidal nevus growth into melanoma: Correlation of cytogenetic features and clinical/imaging risk factors for transformation into melanoma (TFSOM-DIM criteria*)

Cytogenetic features n=86(n (%))				
TCGA Class	Low-risk cytogenetics (TCGA class A and B) n=57	High-risk cytogenetics (TCGA class C and D) n=29	Relative risk (RR) of risk factor being present in high-risk vs low-risk tumours	P value
TCGA A (disomy 3, normal 8q)	39 (45.3)	–	–	–
TCGA B (disomy 3, 8q gain)	11 (12.8)	–	–	–
TCGA A/B (disomy 3, 8q undetermined)	7 (8.1)	–	–	–
Total low-risk	57 (66.3)	–	–	–
TCGA C (monosomy 3, 8q gain<3)	–	15 (17.4)	–	–
TCGA D (monosomy 3, 8q gain≥3)	–	3 (3.5)	–	–
TCGA C/D (monosomy 3, 8q undetermined)	–	11 (12.8)	–	–
Total high-risk	–	29 (33.7)	–	–
Risk factors related to transformation into melanoma (TFSOM-DIM* criteria)				
TFSOM risk factor	Low-risk cytogenetics (TCGA class A and B) n=57	High-risk cytogenetics (TCGA class C and D) n=29	RR of risk factor being present in high-risk vs low-risk tumours	P value
Risk factors present at date of initial presentation (DIP)				
Thickness >2 mm by ultrasonography	30 (52.6)	16 (55.2)	1.05	p=0.696
Presence of subretinal fluid on optical coherence tomography	9 (15.8)	12 (41.4)	2.62	p=0.015
Symptoms of vision ≤20/50 by Snellen visual acuity testing	2 (3.5)	2 (6.9)	1.97	p=0.322
Presence of orange pigment by fundus autofluorescence	7 (12.3)	2 (6.9)	0.56	p=0.084
Tumour hollowness by ultrasonography	25 (43.9)	16 (55.2)	1.26	p=0.220
Diameter >5 mm on fundus photography	45 (78.9)	17 (58.6)	0.74	p=0.011
Total TFSOM-DIM score mean (median, range)	2.1 (2.0, 0.0–6.0)	2.2 (2.0, 0.0–5.0)	n/a	p=0.278
Risk factors present at date of transformation into melanoma (DOT)				
Thickness >2 mm by ultrasonography	50 (87.7)	27 (93.1)	1.06	p=0.186
Presence of subretinal fluid on optical coherence tomography	36 (63.2)	25 (86.2)	1.36	p=0.038
Symptoms of vision ≤20/50 by Snellen visual acuity testing	10 (17.5)	10 (34.5)	1.97	p=0.097
Presence of orange pigment by fundus autofluorescence	23 (40.4)	17 (58.6)	1.45	p=0.276
Tumour hollowness by ultrasonography	46 (80.7)	23 (79.3)	0.98	p=0.386
Diameter >5 mm on fundus photography	54 (94.7)	29 (100.0)	1.06	p=0.991
Total TFSOM-DIM score mean (median, range)	3.8 (4.0, 1.0–6.0)	4.5 (5.0, 3.0–6.0)	n/a	p=0.003
TFSOM-DIM Criteria, remembered by mnemonic 'To Find Small Ocular Melanoma, Doing IMaging': T: thickness >2 mm on ultrasonography; F: presence of subretinal fluid on spectral domain optical coherence tomography; S: visual symptoms of visual acuity of 20/50 or worse by Snellen testing; O: presence of orange pigment on fundus autofluorescence; M: Melanoma hollowness on ultrasonography; DIM: basal diameter greater than 5 mm by fundus photography.				
*TFSOM-DIM criteria adapted from: Shields CL, Dalvin LA, Ancona-Lezama D, <i>et al.</i> Choroidal nevus imaging features in 3,806 cases and risk factors for transformation into melanoma in 2,355 cases: The 2020 Taylor R. Smith and Victor T. Curtin Lecture. <i>Retina</i> 2019;39:1840–1851.				
TCGA, The Cancer Genome Atlas.				

figure 2A). Given the limited sample size, Kaplan-Meier analysis was inconclusive as to whether tumours with fast rates of thickness increase (>0.5 mm/year) metastasise at greater rates (p=0.06, online supplemental figure 3A). We speculate that the aggressive nature of tumours with high-risk cytogenetics, which lead to metastasis and death, also impacts the clinical behaviour of the tumour itself, as growth rates in thickness, growth rates in basal dimension and the number of TFSOM-DIM clinical/imaging risk factors all significantly correlate with a high-risk cytogenetic profile. Importantly, these clinical criteria provide critical information useful for predicting the cytogenetic makeup and behaviour of these tumours once they have undergone malignant transformation.

At our practice, it is standard to offer treatment immediately on diagnosis of melanoma without any recommendation of delay, and biopsy is routinely performed concurrently with treatment. Due to the risk of visual decline after tumour radiation, some groups have explored the prospect of delayed treatment of small choroidal melanomas in efforts to preserve vision. Damato *et al* explored the prospect of delaying treatment of small choroidal melanoma until growth is observed as a means to avoid vision loss as long as possible.¹⁶ Their online prognostication model output suggested that such a short delay in treatment of tumours with growth would not significantly increase the rate of metastasis. Our finding that growth in basal diameter

Table 4 Clinical factors predictive of tumour cytogenetics in 86 patients with choroidal nevus growth into melanoma: Cytogenetic tumour profile based on rates of absolute and percent growth in thickness and basal diameter

Cytogenetic outcomes based on tumour activity – absolute growth rate				
Rate of tumour thickness increase by ultrasonography (mm/year)	Low-risk cytogenetics (TCGA class A and B)	High-risk cytogenetics (TCGA class C and D)	Relative risk (RR) of high-risk cytogenetic profile	P value
≤ 0.5 mm/year thickness increase n=58	43 (74.1)	15 (25.9)	1.93	p=0.027
> 0.5 mm/year thickness increase n=28	14 (50.0)	14 (50.0)		
Rate of tumour basal diameter increase by fundus photography (mm/year)	Low-risk cytogenetics (TCGA class A and B)	High-risk cytogenetics (TCGA class C and D)	RR of high-risk cytogenetic profile	P value
≤ 0.7 mm/year base increase n=58	44 (75.9)	14 (24.1)	2.22	p=0.007
> 0.7 mm/year base increase n=28	13 (46.4)	15 (53.6)		
Cytogenetic outcomes based on tumour activity – percent growth rate				
Rate of tumour thickness increase by ultrasonography (percent growth per year)	Low-risk cytogenetics (TCGA class A and B)	High-risk cytogenetics (TCGA class C and D)	RR of high-risk cytogenetic profile	P value
≤ 20% per year thickness increase n=55	42 (76.4)	13 (23.6)	2.18	p=0.008
> 20% per year thickness increase n=31	15 (48.4)	16 (51.6)		
Rate of tumour basal diameter increase by fundus photography (percent growth per year)	Low-risk cytogenetics (TCGA class A and B)	High-risk cytogenetics (TCGA class C and D)	RR of high-risk cytogenetic profile	P value
≤ 10% per year basal diameter increase n=57	42 (73.7)	15 (26.3)	1.83	p=0.042
> 10% per year basal diameter increase n=29	15 (51.7)	14 (48.3)		

TCGA, The Cancer Genome Atlas.

greater than 10% per year nearly doubled the risk for high-risk cytogenetics aligns with their report in that growth rates among small tumours can certainly be suggestive of underlying metastatic potential. Therefore, conservative management of small tumours, via immediate diagnostic biopsy or initiation of treatment, is optimal. Damato *et al* advised biopsy of the tumour in cases where delayed treatment is being considered to preserve vision. Our group offers immediate treatment when the suspicion of malignant transformation arises.

Currently, the most robust prognostication method for choroidal melanoma is tumour cytogenetic analysis.¹⁷ However, this technology is expensive, and many patients inherently do not have access to it. Also, results can take months to acquire, which can hinder multidisciplinary, collaborative planning surrounding the melanoma diagnosis. Therefore, there is great utility in using readily available clinical information as a surrogate in such cases, as it might aid in patient counselling and expectation setting while giving insight into prognosis. Therefore, we urge ophthalmologists monitoring choroidal nevus to keep consistent records of each of the clinical/imaging results listed above, especially the TFSOM-DIM criteria and changes in tumour thickness and basal diameter. As these data hold predictive value, they can provide important insight to the ocular oncology team when counselling the patient and when deciding what next diagnostic or therapeutic interventions would be optimal. Once suspicion arises of nevus transformation into melanoma, prompt

referral to an ocular oncologist is appropriate, and sharing clinical features recorded while following the nevus may aid in predicting tumour cytogenetics in cases where biopsy may not be immediately possible.

There are inherent limitations that should be recognised in this data analysis. This was a retrospective data collection with a relatively small cohort of 86 patients with documented growth of nevus into melanoma. Diagnosis of the initial nevus was done clinically based on the most up-to-date risk stratification methods at the time; however, these diagnoses were not confirmed via cytopathology, as this was not standard of care. There was one tumour with aggressively rapid growth over 3 months (increase in base size by 8 mm, increase in thickness by 3 mm) that caused our maximum values to skew up to 32 mm/year growth in basal diameter and 12 mm/year growth in thickness. Moreover, due to the advancement of technology and the development of OCT and autofluorescence during the study period, a small proportion of data points prior to these technologies pertaining to SRF and orange pigment used fundus photography as a surrogate. Fortunately, of the 1032 TFSOM-DIM data points collected in this data set, this was only the case for 39 (3.8%) data points. Additionally, fine-needle aspiration biopsy collects a small portion of tumour cells from tumours which may have cellular heterogeneity, which portends a risk of inaccurate cytogenetic analysis in some cases. Despite these features, choroidal nevus transformation into melanoma is a rare event, estimated at 1 in 8845 cases (0.01%) annually, making this cohort invaluable.⁹

CONCLUSION

In conclusion, although transformation of choroidal nevus into melanoma is an uncommon event, the clinical characteristics of these tumours offer crucial information into the likelihood of favourable or unfavourable prognostic cytogenetic characteristics and metastatic outcomes. Choroidal nevus that demonstrates greater rate of increase in thickness (> 0.5 mm/year or $>20\%$ per year) and/or greater rate of increase in basal diameter (> 0.7 mm/year or $>10\%$ per year) will likely have greater rate of expressing high-risk cytogenetics, associated with increased rates for tumour metastasis and death. Furthermore, choroidal nevi which possess more TFSOM-DIM risk factors by DOT are more likely to be characterised by high-risk cytogenetics.

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